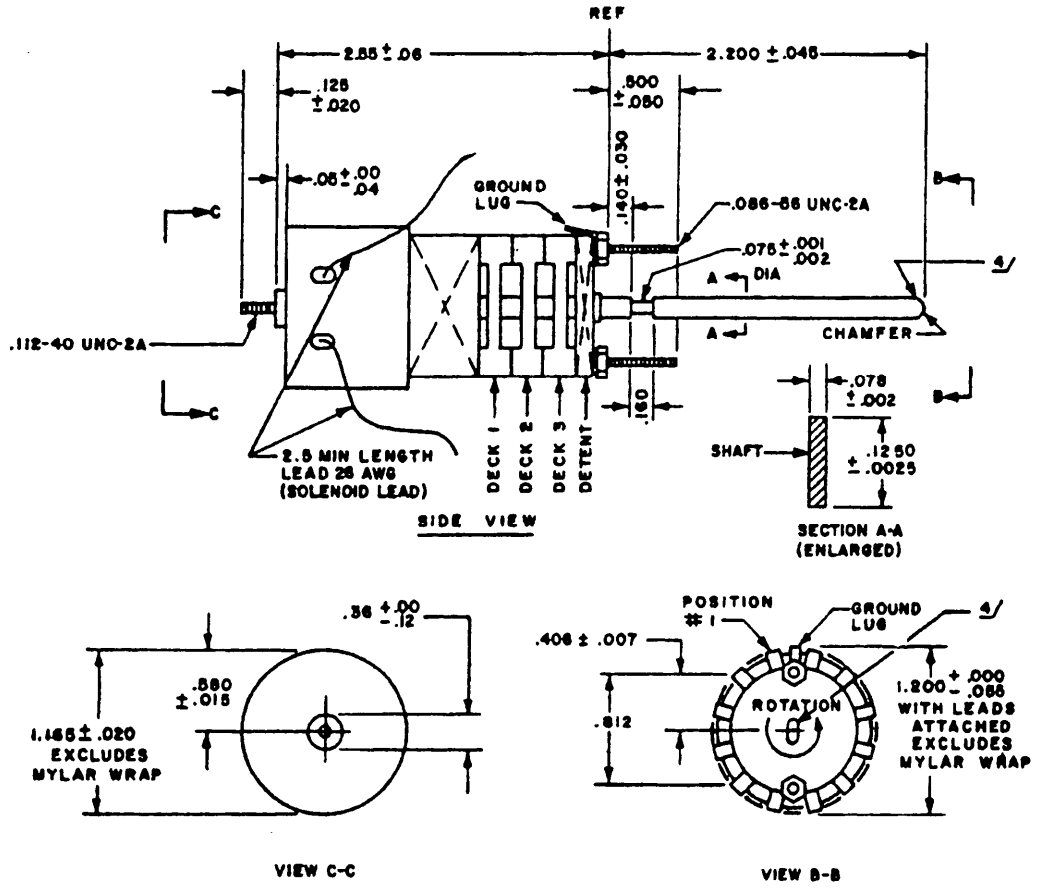
MILITARY SPECIFICATION SHEET

SWITCH, STEPPING, DIRECT CURRENT, ELECTROMECHANICAL, AIRCRAFT, TYPE II (FOR INTERVALOMETER, TYPE DTU-29/A)

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switch described herein shall consist of this document and the latest issue of MIL-S-85242(AS).



NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.010.
3. All components are to be properly assembled to insure smooth functioning.

4/ Switch is shown in the arm position. Scribe notch in top end of shaft.

Example of part number:

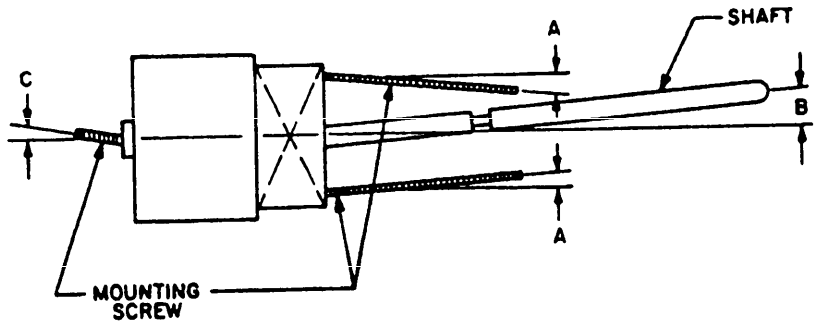
M85242/2-1

- M prefix
- General specification number
- Detail specification number
- Dash number

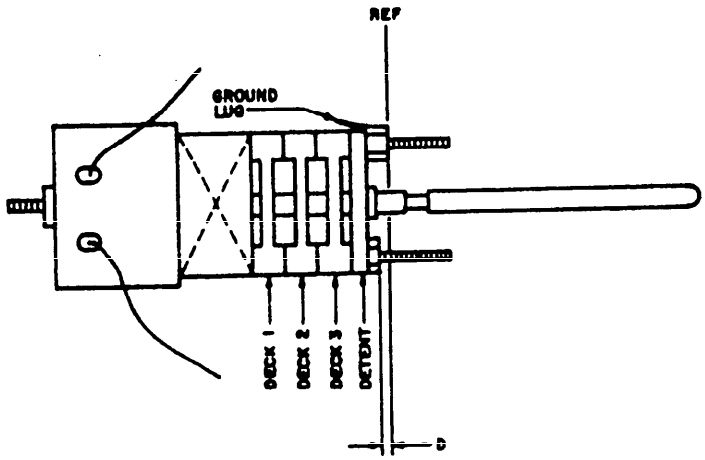
FIGURE 1. Dimensions and configuration.

FSC 5930

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Mounting screw and shaft axial alignment (decks and detents installed but not shown for clarity).



Screw height tolerance (switch decks and detents installed).

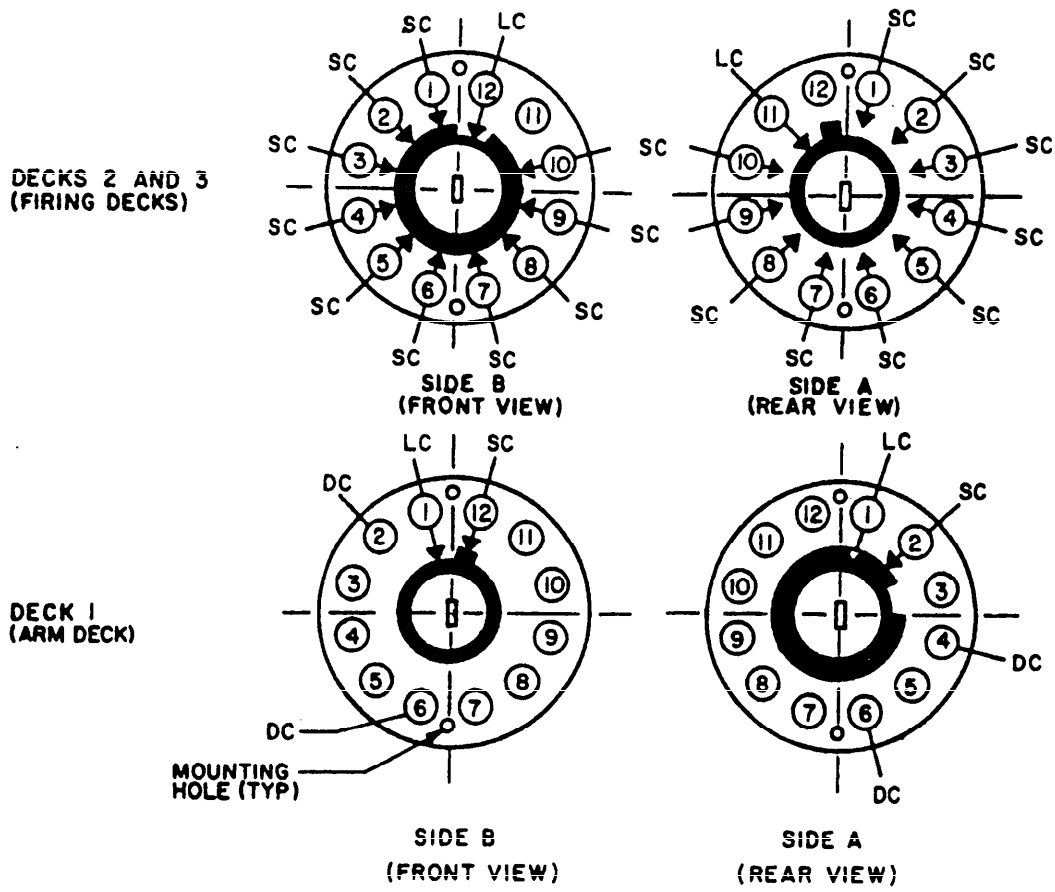
NOTES:

1. Maximum deviation:

   A = ±.010 inch
   B = ±.010 inch
   C = ±.005 inch
   D = ±.005 inch

FIGURE 2. Switch assembly tolerance.

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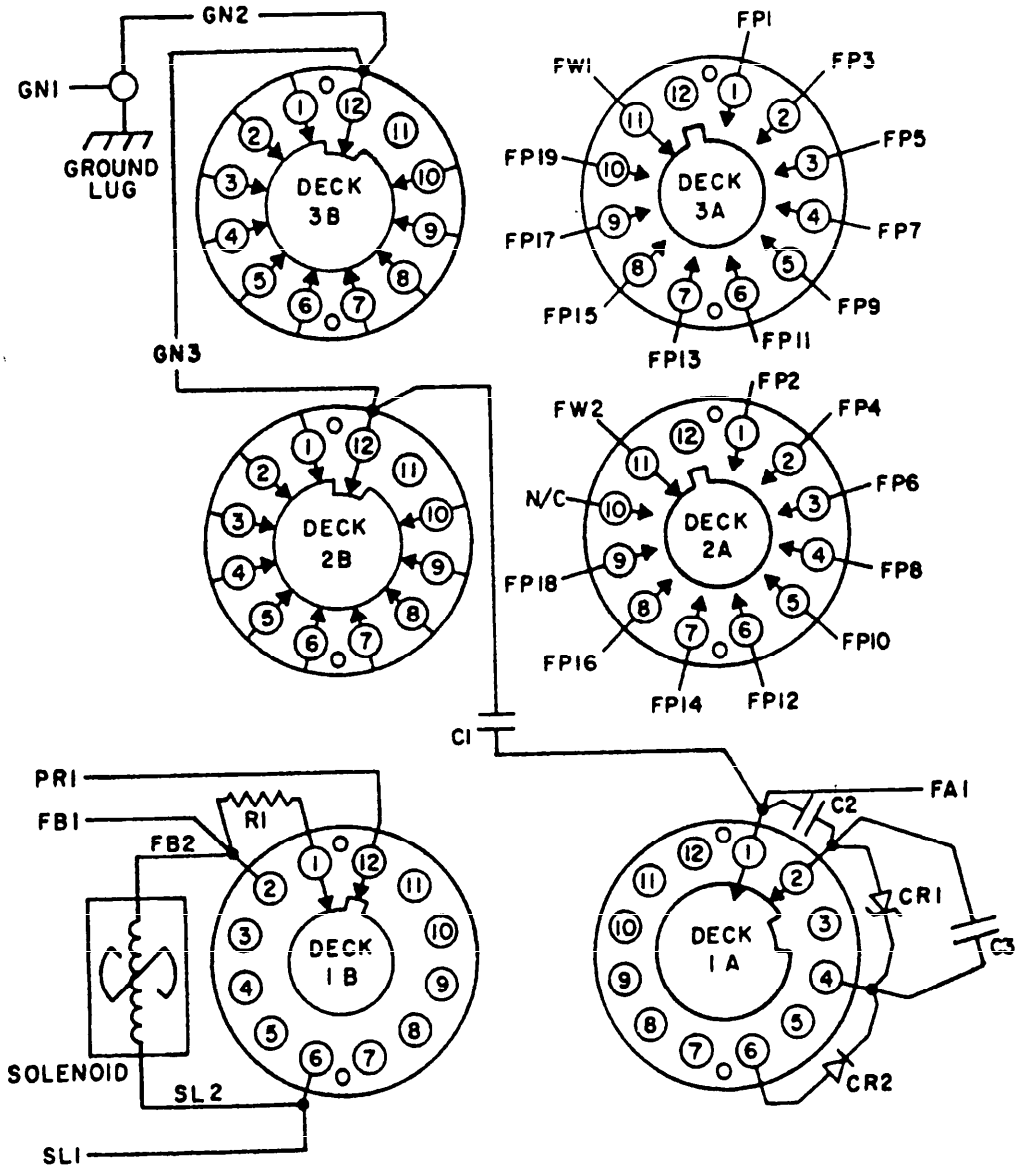


NOTES:

1. Switch is shown in the arm position.
2. Legend:
   DC = Dummy Clip
   LC = Long Clip
   SC = Short Clip

FIGURE 3. Switch decks.

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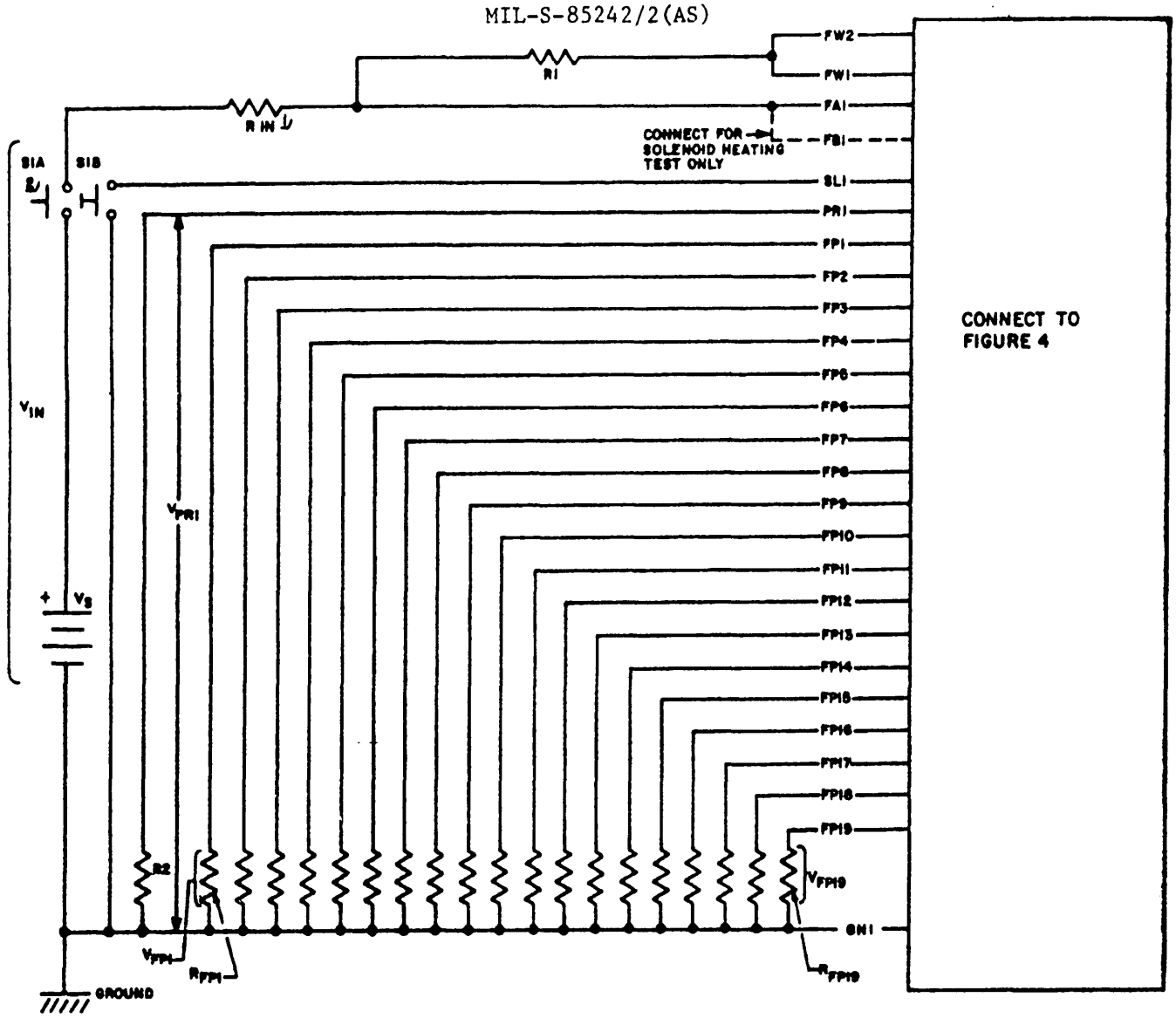


Required Additional Components		
Part Number	Part Name	Ref Des
1N4742A	Semiconductor Device (Zener Diode)	CR1
JAN1N3613	Semiconductor Device (Diode)	CR2
M39014/05-2261	Capacitor (.01 Mfd, 100 Volt)	C1
M39014/02-1350	Capacitor (.1 Mfd, 100 Volt)	C2
CW20C473K	Capacitor (.047 Mfd, 50 Volt)	C3
RCR05G513JS	Resistor (51K Ohm, 1/8 Watt)	R1
1/ 200AS108-26C10	Wire, Stranded	FB2, SL2
QQ-W-343 Type S	Wire, Solid Tinned Copper, AWG 26	GN2, GN3

1/ Part number shown is NAVAIR part number.

FIGURE 4. Switch wiring for type DTU-29/A intervalometer.

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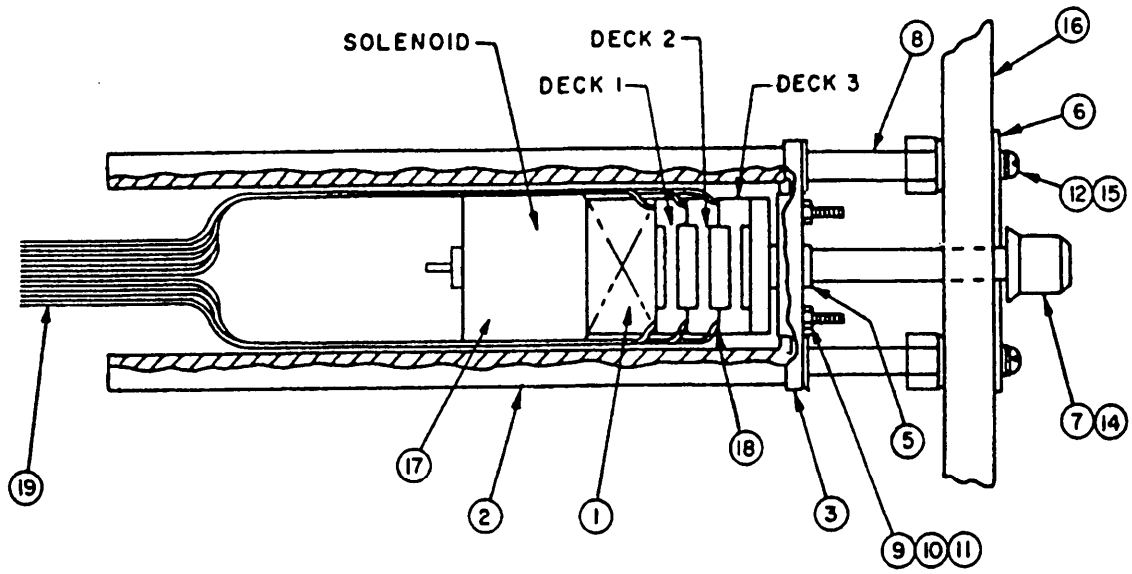
1/ R IN includes line and voltage source resistance.
2/ Switch S1 may be replaced by relays or solid state devices.

NOTES:
1. Test circuit, except stepping switch, shall be maintained at 25° ±10°C for all tests.

Parts List		
Part Number	Part Name	Ref Des
As required	Power Source (10 to 20 Volts DC, 5 Amp)	Vs
As required	Switch (DPST, Normally Open, 5 Amp Contacts)	S1
RER60F2500FR	Resistor (250 Ohm, 5 Watt, 5%)	R1
RCR32G102JS	Resistor (1000 Ohm, 1 Watt, 5%)	R FP (1-19)
RCR07G103JS	Resistor (10K Ohm, 1/4 Watt, 5%)	R2
Input Resistance - Specified in Table II		R IN

FIGURE 5. Electrical test circuit.

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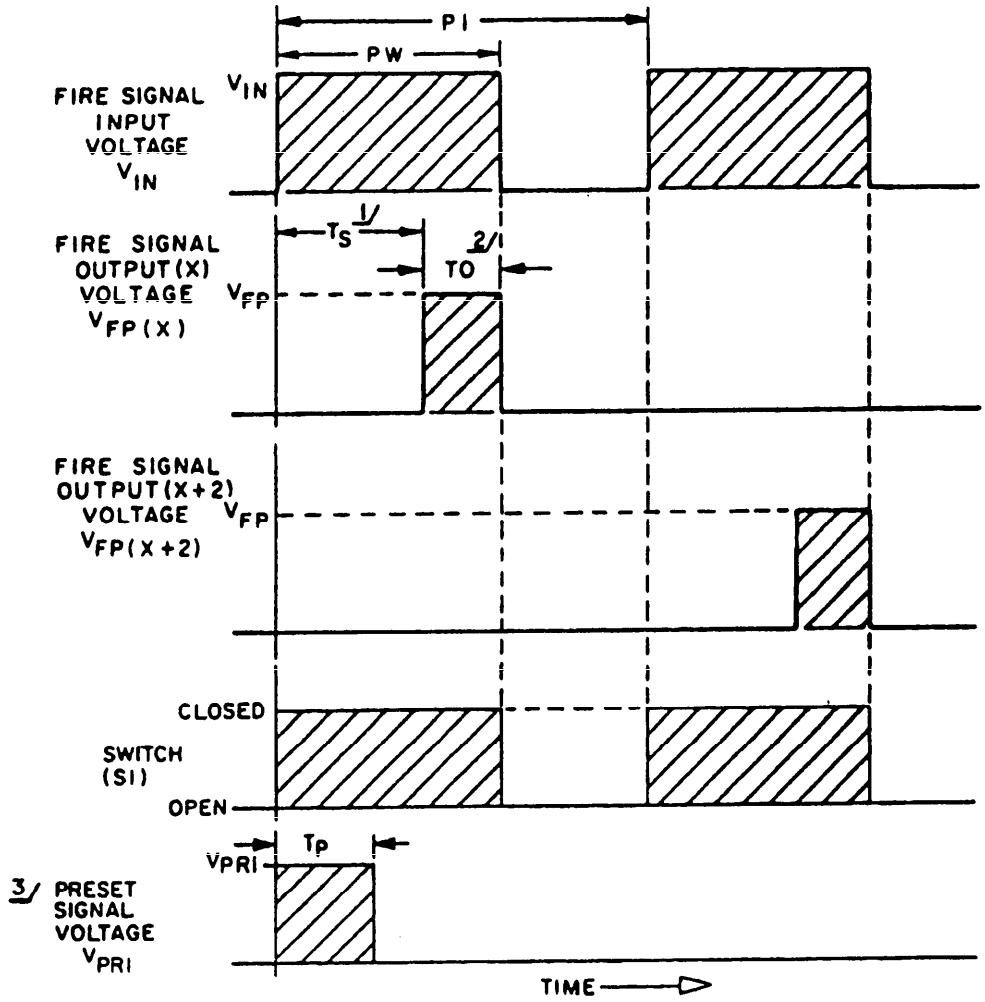


Parts List			
Reqd	Part Number	Part Name	Item
1	M85242/2-1	Switch, Stepping	1
1	1120AS241	Housing, Intervalometer	2
1	1120AS243	Cap, Switch	3
1	1120AS245	Bushing, Shaft	5
1	1120AS246	Plate, Load/Arm	6
1	1120AS248	Knob, Switch	7
3	1120AS249	Stud, Mounting	8
2	MS15795-802	Washer, Flat	9
2	MS35338-134	Washer, Lock	10
2	MS35649-224	Nut, Hex	11
3	MS35207-231	Screw	12
1	MS51021-101	Screw, Set	14
3	MS35335-30	Washer	15
1	Figure 8	Mounting Bulkhead	16
10 Inch	MIL-I-15126, Type MFT-2.5 Width .62	Tape	17
AR	QQ-S-571, SN60 or SN63	Solder	18
AR	200AS108-26C10	Wire	19

1/ Part number shown is NAVAIR part number.

FIGURE 6. Switch mounting for test.

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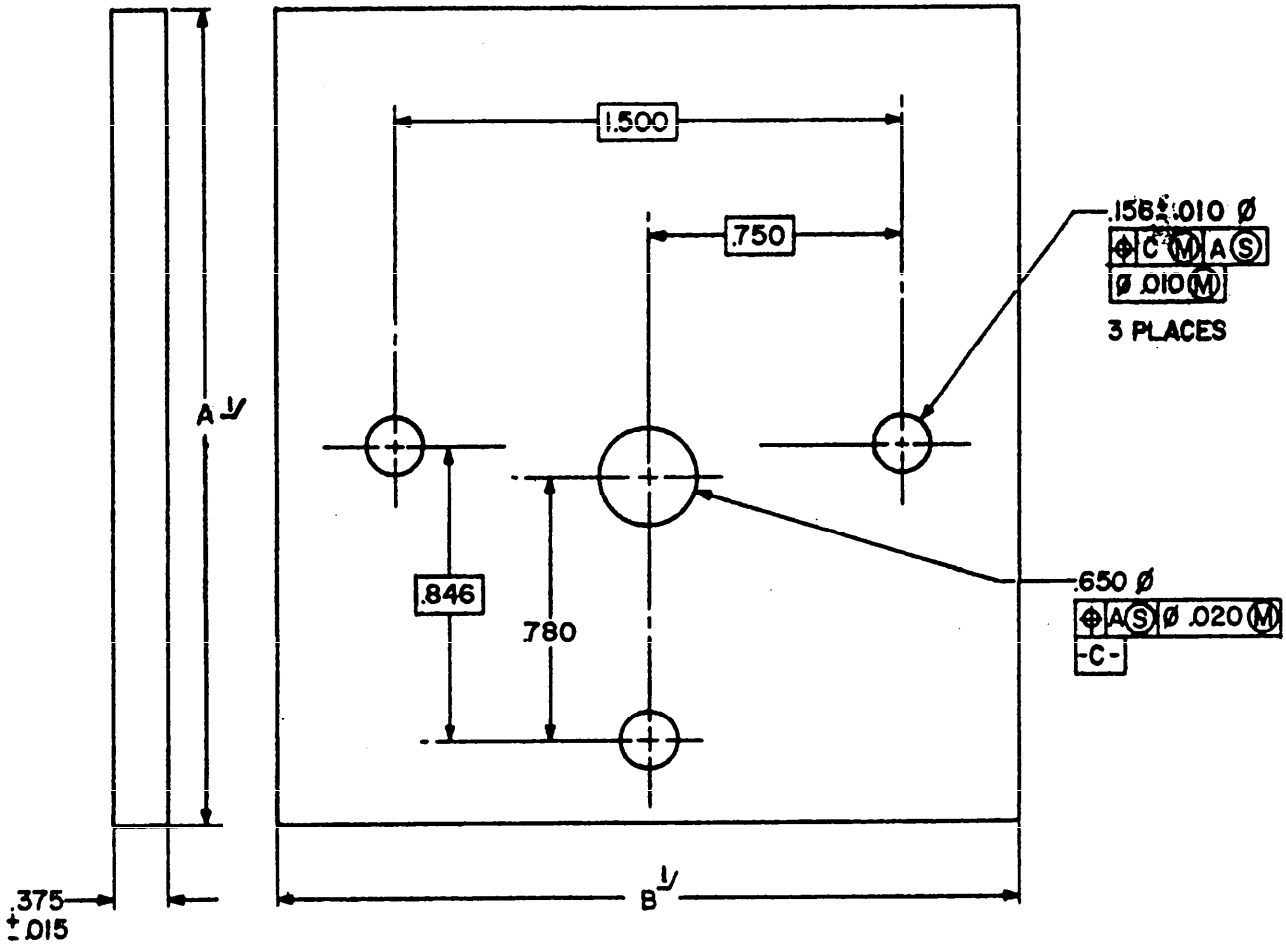
1/ Fire signal output delay (T(S)): 23 milliseconds or less.
2/ T(O): Output (firing) pulse width.
3/ Preset signal (V_{PR1}) is present for first pulse only.

NOTES:

1. Output (X): Any output 1 thru 19 (see figure 5).
2. For parameters, see table II.

FIGURE 7. Stepping switch timing relationship.

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1/ Dimensions: "A" and "B" as required.

NOTE:

1. Material: Aluminum.

FIGURE 8. Mounting bulkhead.

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TABLE I. Summary of switch connections.

Ref Des (Fig 4&5)	Termination From Deck	Termination From Clip	Termination To Deck	Termination To Clip	Wire AWG	Color	Length Inch	Function
FP1	3	1	None		-	-	-	Firing Position #1
FP2	2	1	None		-	-	-	Firing Position #2
FP3	3	2	None		-	-	-	Firing Position #3
FP4	2	2	None		-	-	-	Firing Position #4
FP5	3	3	None		-	-	-	Firing Position #5
FP6	2	3	None		-	-	-	Firing Position #6
FP7	3	4	None		-	-	-	Firing Position #7
FP8	2	4	None		-	-	-	Firing Position #8
FP9	3	5	None		-	-	-	Firing Position #9
FP10	2	5	None		-	-	-	Firing Position #10
FP11	3	6	None		-	-	-	Firing Position #11
FP12	2	6	None		-	-	-	Firing Position #12
FP13	3	7	None		-	-	-	Firing Position #13
FP14	2	7	None		-	-	-	Firing Position #14
FP15	3	8	None		-	-	-	Firing Position #15
FP16	2	8	None		-	-	-	Firing Position #16
FP17	3	9	None		-	-	-	Firing Position #17
FP18	2	9	None		-	-	-	Firing Position #18
FP19	3	10	None		-	-	-	Firing Position #19
FW1	3	11	None		-	-	-	Firing Wire 1
FW2	2	11	None		-	-	-	Firing Wire 2
FA1	1	1	None		-	-	-	Fire Input
FB1	1	2	None		-	-	-	Fire Power
FB2	Solenoid		1	2	26	White	As Reqd	Fire Power
SL1	1	6	None		-	-	-	Solenoid Step
SL2	Solenoid		1	6	26	White	As Reqd	Solenoid Step
PR1	1	12	None		-	-	-	Preset
GN1	Ground Lug		None		-	-	-	Ground
GN2	Ground Lug		3	12	26	Bare	As Reqd	Ground
GN3	3	12	2	12	26	Bare	As Reqd	Ground
GN4	Ground Lug		None		-	-	-	Ground
CR1	1	2	1	4	Diode		As Reqd	RFI Suppression
CR2	1	4	1	6	Diode		As Reqd	RFI Suppression
C1	1	1	2	12	Capacitor		As Reqd	RFI Suppression
C2	1	1	1	2	Capacitor		As Reqd	RFI Suppression
C3	1	2	1	4	Capacitor		As Reqd	RFI Suppression
R1	1	1	1	2	Resistor		As Reqd	RFI Suppression

NOTES:

1. Use wires FA1, FB1, FP1 thru FP19, FW1, FW2, GN1, GN4, PR1 and SL1 for first article inspections and wires FA1, SL1 and GN1 for sample inspections. Use wires FA1, FB1, SL1 and GN1 for solenoid heating test.
2. Cathode ends of diodes CR1 and CR2 are connected to clip 4 of deck 1.

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TABLE II. Switch electrical test.

Test No.	Input Parameters						Minimum Acceptable Output (Each Output in Sequence)		Test Temperature Range 2/ (Deg C)	Remarks
	Voltage (Volts) V IN 1/	Pulse Width (Sec) PW 1/	Pulse Interval (Sec) PI 1/	Line Resistance (Ohms) R IN 1/	Pulse Qty	Initial Switch Position	Final Switch Position	Pulse Width (Sec) T(O) 1/		
1	17.5	0.023	0.050	5.50	11	Arm	Load	-	-55 to +71	
2	17.5	0.023	0.050	1.00	11	Arm	Load	-	-10 to +71	
2A	12.5	0.023	0.050	1.00	11	Arm	Load	-	-55 to -15	
3	17.5	0.035	0.060	5.50	11	Arm	Load	0.010	-55 to +71	
4	17.5	0.035	0.060	1.00	14	Arm	Load	0.010	-10 to +71	
4A	12.5	0.035	0.060	1.00	14	Arm	Load	0.010	-55 to -15	

1/ Tolerance: Resistance R IN = ±2% or 0.05 Ohm (whichever is greater); voltage ±2%; time ±2%.
2/ Temperature range: -55°C to +71°C unless otherwise specified.

NOTES:

1. Switch shall step with each input. Switch shall not electrically advance beyond the safe (load) position (11 steps from arm) except for solenoid heating test.
2. Switch contact bounce: Steady output shall occur within 23 milliseconds at application of step pulse.
3. Abbreviations: Sec - Seconds.
4. Input voltage rise time: 1.5 milliseconds or less (straight line from 10% to 90% of input voltage).
5. Preset voltage (V PR1) must occur coincident with the initial pulse (switch in "ARM" position) and remain for 6.0 to 25 milliseconds. The preset voltage must not occur again until the switch is reset to "ARM" and again energized.
6. For timing relationship see figure 7.

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Requirements:

Physical data:

Dimensions and configuration: See figure 1. The 1.165 ±.020 dimension, excluding mylar wrap and solenoid lead wires, shall be carefully checked. No clips, ground lug, or electronic components shall protrude beyond the 1.200 maximum dimension.

Shaft alignment: See figure 2. Shaft flat shall be aligned parallel to mounting holes during assembly (see figure 3).

Switch clips: Straight (flat).

Marking: Military part number, M85242/2-1, and manufacturer's name or source code, and manufacturer's part number on coil dust cover.

Material:

Deck (wafer section): Diallyl phthalate in accordance with MIL-M-14. Decks shall be smooth and free of flash.
Shaft: Stainless steel.
Switch clips: Silver alloy.
Rotor: Silver alloy.

Rotor Cut: Homing cut on deck 1A; right hand homing tab on deck 1B; decks 2A and 3A shall be shorting and decks 2B and 3B shall be shorting. Switch clips on decks 2 and 3 (positions 1 thru 10) shall contact side A rotor or side B rotor or both in any shaft position.

Design and construction: The switch shall be located within the Type DTU-29/A Intervalometer. It shall consist of a 12 position rotary solenoid ratchet mechanism which shall allow any of 12 shaft positions to be selected electrically (wire FB1 connected to wire FA1) or manually. The switch shall have two firing decks and one arm deck (see figure 3) and shall distribute the incoming firing signal in sequence to each of up to ten electrically-actuated igniters per firing deck.

Design and construction of upper detents:

a. With no torque applied to shaft, upper detents shall positionally retain switch decks in any of twelve 30° increments any time the switch is not energized. Deck positions shall be retained so as to provide full contact between switch common wiper finger and deck clips.

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b. In advancing switch manually, upper detents shall not seat prior to positive ratchet drive engagement (ensuring complete 30° actuation before next application of power) in any of 12 positions. Upper detent seating shall occur within 5° $\pm 3^{\circ}$ after full ratchet engagement.

c. Upper detents and ratchet mechanism shall seat positively with good tactile feedback when operated manually.

Overthrow control: With each application of power applied to the solenoid, the stepping switch assembly shall step, indexing the decks 30°. No double steps, partial steps or intermediate indexing (eg, 45° rotation) shall occur.

Stepper switch and deck design tolerances: When manually moving the switch assembly to arm position (switch position 12), positive ratchet drive engagement shall occur after electrical contact between arm common and the arm clip (clip 2, see figure 3, deck 1, side A). Contact shall occur at 5° $\pm 3^{\circ}$ of rotation before ratchet engagement occurs and 10° $\pm 3^{\circ}$ before upper detent seating.

Operational characteristics:

Safe (load) position: The eleventh step pulse shall advance the switch into the safe (load) position (switch position 11). When wired as shown in figure 5 (wire FB1 disconnected), the switch shall not be capable of being electrically advanced out of the safe (load) position but shall be manually advanced to the arm position.

Arm position: The arm position (switch position 12) shall provide a ground circuit to all firing output circuits and shall allow the switch to be electrically advanced.

Fire positions: Switch positions 1 through 10 are fire positions and shall be sequenced in that order.

Ground circuit: Shall ground all firing output lines except selected firing position(s).

Preset circuit: Shall provide an output which shall exist only when the switch is in the arm position (position 12).

Coil data:

Coil Current and Pickup (operate) voltage: N/A.

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Characteristics: When connected to the test circuit shown in figure 5 and subjected to tests of table II, the dropout voltage and duty cycle shall be as follows:

a. Dropout voltage: 1.0 VDC minimum (0.5 ohm line resistance over temperature range -55°C to 71°C).

b. Duty cycle: The duty cycle is 10 percent. Maximum continuous on time is 35 milliseconds.

Solenoid heating: The switch shall be mounted as shown in figure 6. The test circuit of figure 5 shall be used except that wire FB1 shall be connected to wire FA1. Test 4 of table II shall be used at room temperature for 120 continuous steps (pulses). The switch shall function properly before and after this test.

Solenoid:
Size: Size 2 with dust cover.
Type: Direct drive ratchet type. The ratchet return spring and clutch spring shall operate freely with no interference between springs or any other components.
Wrap: One to two turns of MIL-I-15126, type MFT-2.5 tape, .62 inches wide (see figure 6, item 17).
Leads: AWG 26 (see figures 1 and 4 and table I), oriented as shown in figure 1.

Contact data (switch decks):

Overload: N/A.

Overload resistive: N/A.

Overload carrying: N/A.

Contact rating: 800 revolutions

Make: 0.25 amp non-inductive (at 30 volts) on decks 1, 2 and 3.

Break: 0.25 amp non-inductive (at 30 volts) on decks 1B, 2 and 3; solenoid homing current on deck 1A.

Carry: 5.0 amp non-inductive on decks 2 and 3; 5.0 amp inductive on deck 1A; 1.0 amp non-inductive on deck 1B.

Current range: Contact carry current shall range between zero and 5.0 amp.

Lubricant: Wm. F. Nye, number 813.

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Electrical data:

Interrupter contacts operate and release time: N/A.

Homing time: N/A.

Self-interrupted stepping speed: N/A.

Overvoltage: N/A.

Externally-pulsed stepping speed: The tests of table II shall be performed at each of the following temperatures: 71° ±2°C, 23 ±5°C and -55° ±2°C. The test circuit of figure 5, maintained at 25° ±10°C, wiring per figure 4, and mounting per figure 6 shall be utilized. The switch shall be capable of following an essentially square wave pulse (see figure 7) having a minimum on time of 23 milliseconds and a minimum off time of 25 milliseconds over temperature range -55°C to 71°C. Deck 2 and 3 contact closure shall occur within 23 milliseconds of pulse application.

Dielectric withstanding voltage (at atmospheric pressure):

	Vrms
Between solenoid coil and case	600
Between adjacent, unwired wafer contacts	750

NOTE: Do not wire switch as shown in figures 4 and 5 or table I and do not use table II or mount as shown in figure 6 for dielectric withstanding voltage test.

Dielectric withstanding voltage (at reduced barometric pressure): N/A.

Switch deck: Sides A and B of switch deck shall be electrically connected.

Wire:
- Size: 26 AWG (see figure 4, Ref Des GN2 and GN3). See coil data for solenoid leads.
- Type: Copper
- Insulation: None

Ground circuit resistance (switch in safe or arm position).
- Initial: 0.070 ohm max (after lubrication).
- After 800 operations (5.0 amperes carry): 0.10 ohm maximum all firing output lines to ground.

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Starting torque test: The switch shall be wired as shown in figure 5 (wires FA1, SL1 and GN1 connected) and mounted as shown in figure 6. Using Test No. 3 of table II the starting torque at the shaft of the assembled switch shall be measured by suitable means at a temperature of 25° ±10°C. The starting torque shall be greater than 0.50 inch-pounds (8.0 inch-ounces).

Environmental data:

Salt spray (corrosion):	N/A.
Seal:	N/A.
Terminal strength:	N/A.
Humidity resistance:	N/A.
Corona:	N/A.
Electromagnetic compatibility:	N/A.

Thermal shock: MIL-STD-202, Method 107, test condition A-1, except that the number of cycles shall be 11 and the maximum temperature shall be 71°C. The tests listed in table II shall be conducted prior to the first cycle and upon completion of the final cycle. The first five cycles shall be run continuously with a time of 1 hour for steps 1 and 3. During the next cycle, the switch shall be held at -55°C for 8 hours minimum and the tests in table II shall be conducted with temperature maintained. The test shall continue to and terminate at step 3 with the temperature remaining at 71°C for 8 hours minimum. The tests in table II shall be repeated with the temperature maintained. The first five cycles of the series shall be repeated for the conclusion of the series.

Vibration: MIL-STD-810, Method 514.2, procedure IIA. During a resonant dwell and at the end of vibration in each axis, tests 3 and 4 of table II shall be performed on the switch. Immediately upon completion of the environmental conditioning, the sample shall be visually examined for damage or defects.

Shock: MIL-STD-810, Method 516.2, procedure IV, figure 516.2-2, amplitude a, duration c. The switch shall be rigidly mounted to the shock machine table. For the environmental exposure, one switch shall be placed in the safe position with power applied, and the other switch shall be placed in the arm position with no power applied. Three shocks in each direction shall be applied along three mutually perpendicular axes of the mounting frame (total of 18 shocks). Any inadvertent advance of the switch or malfunction shall be cause for rejection. Vibration isolators shall not be used.

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High and low temperature operation: Part of thermal shock test.

Acceleration: MIL-STD-810, Method 513.2, procedure II, table 513.2-II. The switch shall be tested in three mutually perpendicular axes.

Note: For environmental testing, switch assembly shall be wired as shown in figure 4, and shall be mounted as shown in figure 6. The test circuit of figure 5 shall be used and maintained at 25° ±10°C. The tests in table II shall be conducted during and after being subjected to the following environmental tests: vibration, shock, thermal shock, and acceleration.

Life test: The switch assembly shall be wired as shown in figure 4 and mounted as shown in figure 6. The switch shall be subjected 200 times each to test numbers 1, 2, 3 and 4 of table II at room temperature for a total of 800 tests. The test circuit of figure 5 shall be used and maintained at 25° ±10°C for all tests.

Custodians:
Navy - AS

Review activity:
DLA - ES

Preparing activity:
Navy - AS
(Project 5930-N559)

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